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NOTES on Continuity, Inverse Functions, and Weierstrass' Theorem

These concepts explain why mathematical models behave predictably: small changes in an input produce small changes in an output, measurements can sometimes be reversed to recover unknown quantities, and engineering quantities attain maximum and minimum values over finite operating ranges.

1. Continuity

1.1 Informal meaning

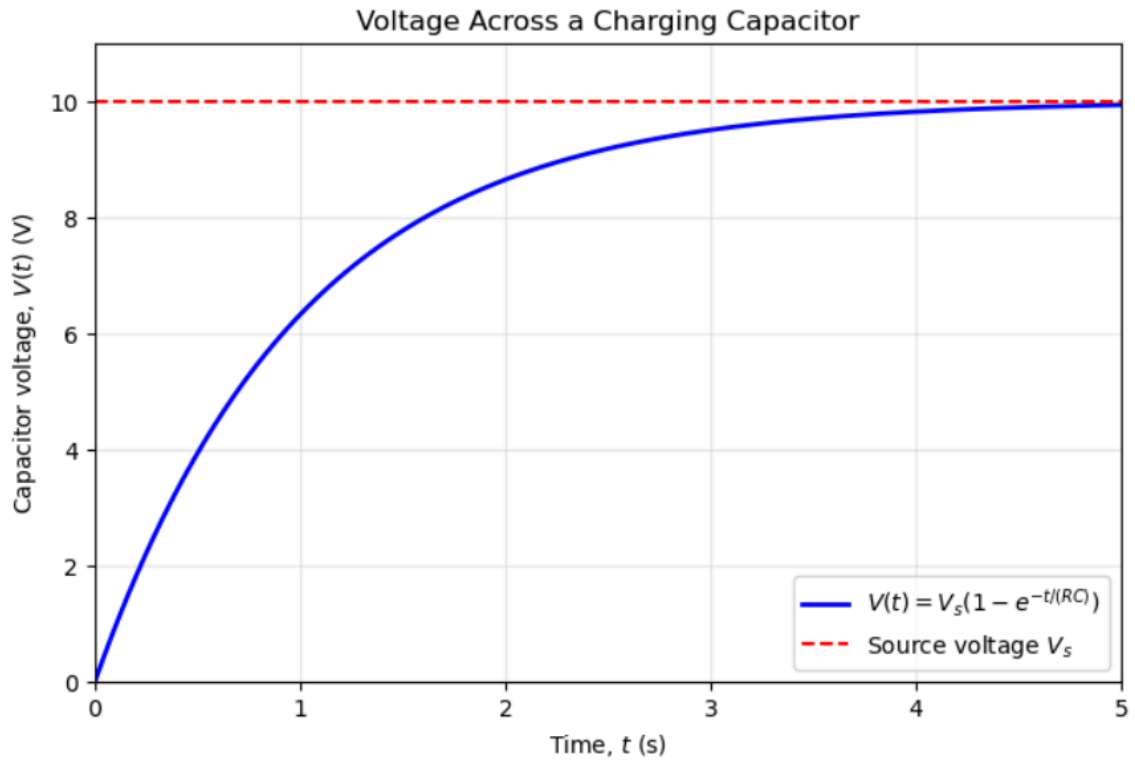
A function is continuous if its graph has no breaks, jumps, holes, or vertical asymptotes within the region being considered.

In engineering terms, if $y = f(x)$ is continuous near $x = a$, then a sufficiently small change in the input x produces a small change in the output y.

For example, the voltage across a charging capacitor,

$$V(t) = V_s \left(1 - e^{-t/(RC)}\right),$$

is continuous for $t \geq 0$. The model predicts that the voltage changes smoothly rather than jumping instantaneously (see plot below).



1.2 Continuity at a point

A function f is continuous at $x = a$ if

$$\boxed{\lim_{x \to a} f(x) = f(a)}.$$

This condition contains three requirements:

1. $f(a)$ is defined.
2. $\lim_{x \to a} f(x)$ exists.
3. The limit equals the function value.

Equivalently,

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a).$$

1.3 Continuity on an interval

A function is continuous on an open interval (a, b) if it is continuous at every point of the interval.

It is continuous on the closed interval $[a, b]$ if:

- it is continuous at every point of (a, b);
- it is right-continuous at a:

$$\lim_{x \to a^+} f(x) = f(a);$$

- it is left-continuous at b:

$$\lim_{x \to b^-} f(x) = f(b).$$

The one-sided conditions are used because the function need not be defined outside $[a, b]$.

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2. Algebra of continuous functions

If f and g are continuous at $x = a$, then the following functions are also continuous at a:

$$f + g, \qquad f - g, \qquad cf, \qquad fg,$$

and

$$\frac{f}{g},$$

provided that $g(a) \neq 0$.

Important to know: Polynomial functions are continuous everywhere. Rational functions are continuous wherever their denominators are nonzero. Exponential, logarithmic and trigonometric functions are continuous on their domains.

Composition theorem

If g is continuous at a, and f is continuous at $g(a)$, then the composite function

$$(f \circ g)(x) = f(g(x))$$

is continuous at a.

Consequently,

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right),$$

provided the relevant continuity conditions hold.

For example,

$$T(t) = 20 + 50e^{-0.2t^2}$$

is continuous because it is formed by composing and combining continuous functions.

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3. Types of discontinuity

Removable discontinuity

Consider

$$f(x) = \frac{x^2-4}{x-2}, \qquad x \neq 2.$$

For $x \neq 2$,

$$f(x) = x + 2.$$

Therefore,

$$\lim_{x\to 2} f(x) = 4,$$

but $f(2)$ is undefined. The discontinuity can be removed by defining $f(2) = 4$.

Jump discontinuity

If

$$f(x) = \begin{cases} f_1(x), & x < a, \\ f_2(x), & x \geq a, \end{cases}$$

and

$$\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x),$$

then $f(x)$ has a jump discontinuity at a. Such discontinuities commonly appear in switching and control models.

Infinite discontinuity

The function

$$f(x) = \frac{1}{x-a}$$

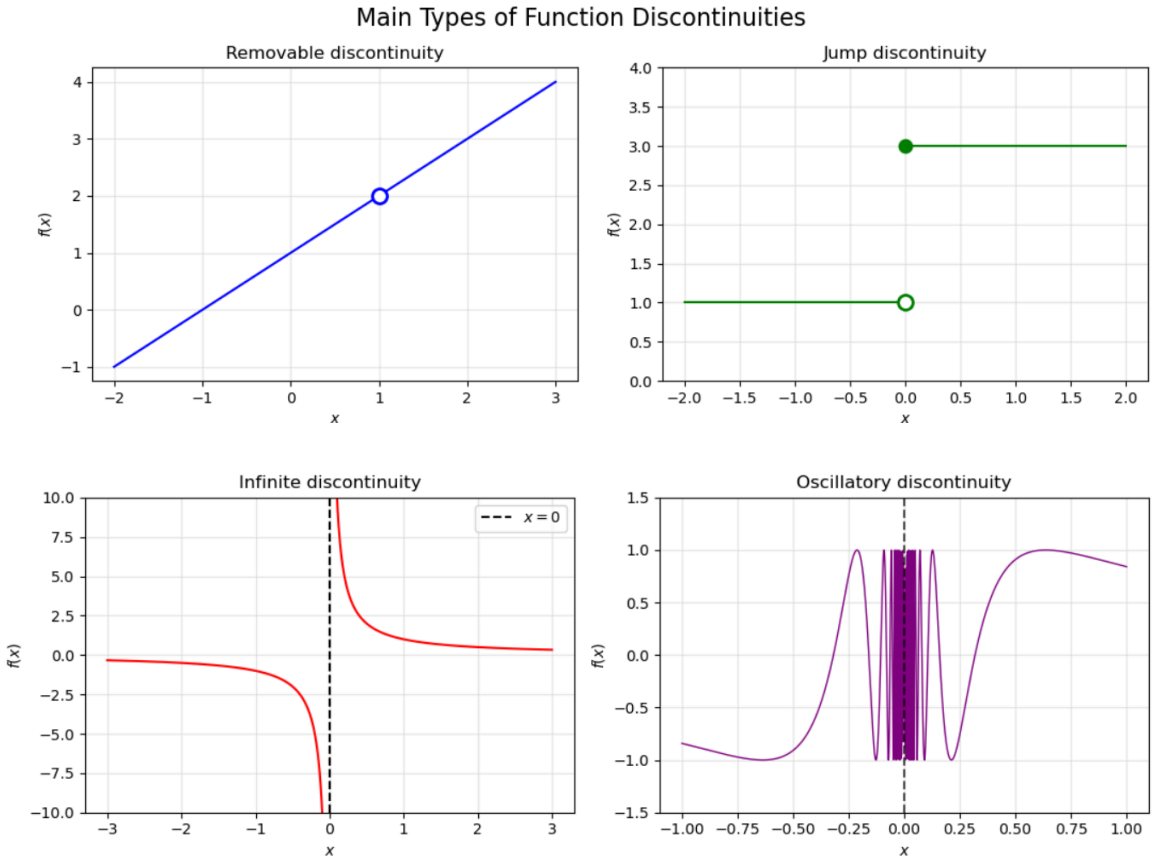
has an infinite discontinuity at $x = a$. The line $x = a$ is a vertical asymptote.

Oscillatory discontinuity

The function

$$f(x) = \sin\left(\frac{1}{x}\right)$$

oscillates indefinitely as $x \to 0$, so the limit does not exist.



4. Inverse functions

4.1 Meaning of an inverse

Suppose

$$y = f(x).$$

An inverse function reverses the input-output relationship:

$$x = f^{-1}(y).$$

The defining properties are

$$f^{-1}(f(x)) = x$$

and

$$f(f^{-1}(y)) = y.$$

The notation $f^{-1}(x)$ does **not** usually mean $1/f(x)$:

$$\boxed{f^{-1}(x) \neq \frac{1}{f(x)}}.$$

For example, if

$$f(x) = 3x + 2,$$

then solving $y = 3x + 2$ for x gives

$x = \frac{y-2}{3}$.

Hence,

$f^{-1}(y) = \frac{y-2}{3}$.

4.2 When does an inverse function exist?

A function has an inverse on its domain if it is **one-to-one**:

$$f(x_1) = f(x_2) \quad \Longrightarrow \quad x_1 = x_2.$$

Graphically, a one-to-one function passes the horizontal-line test: every horizontal line intersects its graph at most once.

A continuous function that is strictly increasing or strictly decreasing on an interval is one-to-one there and therefore has an inverse.

For example, $f(x) = x^2$ is not one-to-one on $\mathbb{R}$, because

$f(-2) = f(2) = 4.$

However, if its domain is restricted to $[0, \infty)$, then it is strictly increasing and

$f^{-1}(x) = \sqrt{x}.$

4.3 Continuity of the inverse function

A useful theorem states:

> If f is continuous and strictly monotonic on an interval I, then f is one-to-one, its range $J = f(I)$ is an interval, and its inverse
>
> $$f^{-1} : J \to I$$
>
> is continuous.

The strict-monotonicity condition means that f is either strictly increasing or strictly decreasing.

Geometrical interpretation

The graph of f^{-1} is obtained by reflecting the graph of f across the line

$y = x.$

If a continuous, strictly monotonic graph has no breaks, its reflection also has no breaks.

4.4 Engineering interpretation

Suppose a sensor output is related to temperature by

$V = f(T).$

If f is continuous and strictly increasing, then:

- each temperature produces exactly one voltage;
- each voltage in the sensor's range corresponds to exactly one temperature;
- the calibration can be inverted:

$$T = f^{-1}(V);$$

- small changes in the measured voltage produce small changes in the recovered temperature.

Continuity alone is not sufficient for invertibility. The calibration curve must also be one-to-one.

Example: Inverting a sensor model

Suppose a sensor follows

$$V(T) = 2 + 0.01T^2, \qquad T \geq 0.$$

This function is *continuous and strictly increasing* for $T \geq 0$. *Therefore, it has a continuous inverse*.

Solve for T:

$$V - 2 = 0.01T^2,$$

so

$$T^2 = 100(V - 2).$$

Because $T \geq 0$,

$$\boxed{T = V^{-1}(V) = 10\sqrt{V - 2}}, \qquad V \geq 2.$$

Here, it is clearer to name the inverse calibration function $T(V)$:

$$\boxed{T(V) = 10\sqrt{V - 2}}.$$

For $V = 11$ V,

$$T = 10\sqrt{11 - 2} = 30.$$

Thus, the measured temperature is 30°C.

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5. Intermediate Value Theorem

The Intermediate Value Theorem provides an important consequence of continuity.

> If f is continuous on $[a, b]$, and L lies between $f(a)$ and $f(b)$, then there exists at least one $c \in [a, b]$ such that
>
> $$f(c) = L.$$

In particular, if

$$f(a)f(b) < 0,$$

then $f(a)$ and $f(b)$ have opposite signs, so there exists at least one $c \in (a, b)$ for which

$$f(c) = 0.$$

The theorem guarantees existence, but not uniqueness.

Engineering meaning

If the temperature of a *continuously operating system* rises from $40^\circ\mathrm{C}$ to $80^\circ\mathrm{C}$, then it must pass through every intermediate temperature, including $60^\circ\mathrm{C}$.

This conclusion would not necessarily hold for a discontinuous model.

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6. Weierstrass' Extreme Value Theorem

In introductory calculus, "Weierstrass' Theorem" usually refers to the **Extreme Value Theorem**.

> If f is continuous on a closed and bounded interval $[a, b]$, then f attains both an absolute maximum and an absolute minimum on that interval.

Thus, there exist points $x_{\min}, x_{\max} \in [a, b]$ such that

$$f(x_{\min}) \le f(x) \le f(x_{\max})$$

for every $x \in [a, b]$.

Notice the word **attains**. The theorem says that the maximum and minimum are actual function values, not merely values that the function approaches.

Why the hypotheses matter

Both conditions are important:

1. The function must be continuous.
2. The interval must be closed and bounded.

Open interval

Consider

$$f(x) = x, \qquad 0 < x < 1.$$

The function is continuous, but the interval is not closed. The function approaches 0 and 1, but it attains neither value. Therefore, it has no absolute maximum or minimum on $(0, 1)$.

Discontinuous function

Consider

$$f(x) = \begin{cases} x, & 0 \le x < 1, \\ 0, & x = 1. \end{cases}$$

The domain $[0, 1]$ is closed and bounded, but f is discontinuous at $x = 1$. Its values approach 1, but no point gives $f(x) = 1$. Thus, it has no absolute maximum.

The hypotheses are sufficient conditions. A function that fails one hypothesis may still have extrema, but the theorem no longer guarantees them.

Finding absolute extrema in practice

For a function that is continuous on $[a, b]$:

1. Find the critical points inside (a, b), usually by solving

$$f'(x) = 0$$

(wait till next week, when we are going to introduce derivatives!)

2. Evaluate f at every critical point.
3. Evaluate $f(a)$ and $f(b)$.
4. Compare all values.

The largest value is the absolute maximum, and the smallest is the absolute minimum.

Weierstrass' theorem guarantees that these extrema exist. Differential calculus helps us locate them.

How important the Weierstrass theorem can be for engineering students?

Weierstrass' Extreme Value Theorem is very important in engineering—not mainly as a computational technique, but as an **existence guarantee**.

If an engineering performance function f is continuous and the allowable design set is closed and bounded, then an optimal design actually exists:

$$\boxed{\text{continuous model} + \text{closed bounded feasible set} \Longrightarrow \text{attained maximum and minimum.}}$$

This principle underlies structural optimization, energy management, control, circuit design, thermal analysis, materials selection and numerical optimization. Before trying to calculate an optimum, an engineer should ask whether an optimum is guaranteed to exist.

For a continuous function $f : [a, b] \to \mathbb{R}$, Weierstrass' theorem states that there are $x_{\min}, x_{\max} \in [a, b]$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for every $x \in [a, b]$.

The theorem guarantees **existence**, but does not tell us:

- where the extrema occur;
- whether they are unique;
- how to calculate them;
- whether a local extremum is also global.

Derivatives, numerical methods and comparisons of candidate values are needed for those tasks.

Main concepts to remember

- Continuity means:

A function f is continuous at $x = a$ if

$$\lim_{x \to a} f(x) = f(a).$$

This condition contains three requirements:

1. $f(a)$ is defined.
2. $\lim_{x \to a} f(x)$ exists.
3. The limit equals the function value.

Equivalently,

$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a).$

- Continuous functions preserve smooth input-output behaviour.
- A function must be one-to-one to possess an inverse.
- A continuous, strictly monotonic function on an interval has a continuous inverse.
- The Intermediate Value Theorem guarantees that continuous functions take every intermediate value.
- Weierstrass' theorem guarantees that a continuous function on a closed, bounded interval attains an absolute maximum and minimum.
- Weierstrass guarantees that extrema exist; derivative methods help locate them.

A deeper engineering interpretation

Weierstrass' theorem supports three important engineering statements:

1. **An optimal design exists.**
   If the objective is continuous and the feasible design set is compact, the best design is not merely approached—it is attained.
2. **A worst-case operating condition exists.**
   A continuous stress, temperature, error or vibration function on a bounded operating range reaches its largest value.
3. **Numerical optimization is meaningful.**
   The theorem justifies searching for an actual optimum, although it does not guarantee that a particular numerical algorithm will find it.

In several variables, the engineering form is:

> If $f : \mathbb{R}^n \to \mathbb{R}$ is continuous and the feasible set $K \subset \mathbb{R}^n$ is nonempty and compact, then f attains a maximum and minimum on K.

In finite-dimensional engineering design, "compact" usually means that the feasible set is **closed and bounded**. This higher-dimensional version is one of the basic theoretical foundations of constrained optimization.

The Notes presented above have been adapted for the Math I (Fall 2026) course delivered at the Tashkent University of Technology, from Prof. G. Neofotistos book (in preparartion) titled Calculus for the 21st-century Engineer - Mathematics, Computation, and Artificial Intelligence (AI). This work is licensed under a Creative Commons "Attribution-NonCommercial-NoDerivatives 4.0 International" license.